Activity 3: Design and Commission an EPA for Orion Optics

Mapping: K3 · K4 · A1
Objective: Design an EPA that controls people, place, product and process, then define commissioning evidence.

Scenario

Orion Optics is moving laser-diode assembly into a small shared lab. The proposed bench has a mat and wrist strap but no documented boundary, common point, insulator policy or readiness test.

Roles

- ESD engineer
- facilities engineer
- lab manager

Detailed procedure

1. Define the ESDS items, process steps and product-sensitivity information required before design approval.
2. Draw the EPA boundary, entry point, product flow and locations of exposed handling.
3. Select the facility/equipment ground reference and show the common-point architecture.
4. Bond the mat, fixtures and other conductive objects on the diagram; flag unresolved isolated conductors.
5. Choose personnel grounding for seated and mobile work and define the pre-use response to failure.
6. Inventory essential/non-essential insulators and select removal, separation, shielding or ionisation controls.
7. Specify storage and transport packaging inside and outside the EPA.
8. Define signage, access, garments, tools and housekeeping requirements.
9. Create a commissioning test plan for ground continuity, worksurface, personnel grounding and ionisation.
10. Assign owners and record locations for qualification and verification evidence.
11. Run a red-team review using the checklist and close every critical gap.
12. Issue the EPA readiness certificate only when all acceptance criteria are met.

Questions

- What makes the area an EPA rather than simply a bench with a mat?
- Which item cannot be made safe by grounding alone?
- What evidence authorises production start?

Evidence to submit

- EPA layout
- Grounding/bonding diagram
- Insulator-control register
- Commissioning plan and readiness certificate

Acceptance criteria

- [ ] Boundary and product flow are unambiguous
- [ ] All conductors have a resolved status
- [ ] Essential insulators have controls
- [ ] Commissioning includes method, limit, owner and response

Debrief

Explain which evidence changed your first assumption, which control is most reliable, and what follow-up measurement is needed.